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Indium-Free Silicon Heterojunction Solar Cells: Achieving High Conductivity With Magnetron-Sputtered SnO_2

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ABSTRACT

Although magnetron sputtering (MS) is widely used for depositing transparent conducting oxide (TCO) films, achieving highly conductive SnO_2 at low process temperatures ($\leq$200°C) remains challenging. This study investigated the effects of substrate temperature, O_2 flow ratio, and residual water vapor on the structural and electrical properties of MS-deposited SnO_2, with a focus on amorphization and H incorporation. Under moderately O_2-deficient conditions, SnO_2 films remained mostly amorphous and exhibited high conductivities (resistivity as low as 1.2×10^{-3} Ωcm) possibly due to the formation of shallow donor states in the disordered network and the reduced contribution of grain boundary scattering in amorphous-dominant films. H, introduced by residual water vapor in the sputtering atmosphere, may play a secondary and indirect role by relaxing the amorphous network or passivating deep-level defects, thereby contributing to improved transport properties. Optimized amorphous SnO_2 films were applied to the front and rear electrodes in Si heterojunction solar cells, achieving 21.1% efficiency. In comparison, devices with Sn-doped In_2O_3 and reactive plasma-deposited SnO_2 demonstrated 22.1% and 22.5% efficiencies, respectively. MS-deposited SnO_2 devices exhibited slightly lower short-circuit current densities and higher contact resistances at the interface with *n*-type hydrogenated nanocrystalline silicon oxide. The former is attributed to sub-bandgap and free-carrier absorption in the MS-SnO_2 films, while the latter arises from a slightly higher work function of MS-SnO_2 and process-induced interfacial degradation during sputtering. These results highlight the need for precise control of structure, H content, and electronic states to advance low-temperature processed, SnO_2-based TCOs for high-efficiency photovoltaics.

1 | Introduction

Rapid expansion of photovoltaic (PV) technologies demands the development of highly efficient and sustainable solar cells to meet the increasing global energy needs [1]. In this regard, Si heterojunction (SHJ) solar cells have emerged as a leading next-generation technology, offering high power conversion efficiencies, bifacial operations, and compatibilities with tandem architecture such as perovskite/Si. Transparent conductive oxide (TCO) is a key component of these devices, which serves as both the front and rear electrodes, facilitating effective charge extraction while maintaining high optical transparency. Sn-doped In_2O_3 (ITO) and emerging high-mobility In_2O_3:H derivatives [2–8] (e.g., In_2O_3:W,H; In_2O_3:Ga,Ti,Zr,H; and In_2O_3:Ce,H) exhibit excellent optoelectronic performances. However, their reliance on In, a critical raw material with limited availability and increasing global demand, presents a significant challenge to their long-term

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scalabilities and cost stability [9]. This resource constraint has stimulated extensive efforts to develop In-free TCO alternatives that are abundant, process-compatible, and stable under PV operating conditions.

Among these candidates, tin dioxide (SnO_2) is exceptional because of its earth abundance, wide optical bandgap, and outstanding chemical stability. Although SnO_2-based TCOs, for instance, F-doped SnO_2 and Sb-doped SnO_2, have been extensively used on glass substrates, their electrical performances at the low process temperatures ($\leq 200°C$) required for temperature-sensitive absorber layers have generally been insufficient. Consequently, their application in high-efficiency SHJ solar cells—where TCO must be directly deposited on hydrogenated amorphous Si (a-Si:H) or nanocrystalline Si (nc-Si:H) without degrading the passivation layer—has been limited. Nevertheless, with the increasing demand for In-free alternatives, recent studies have reexamined the low-temperature fabrication of SnO_2 for SHJ applications [10–15], an area long overlooked due to possible performance limitations. Our previous study has demonstrated fully amorphous SnO_2 (a-SnO_2) thin films synthesized by reactive plasma deposition (RPD), a physical vapor deposition technique capable of forming high-quality films at low temperatures [16]. RPD generates highly ionized species with relatively low kinetic energies (~10–40 eV), thereby minimizing defect production during film growth [17]. Using this approach, we have achieved a-SnO_2 films with resistivities below $1 \times 10^{-3}\,\Omega$cm and exhibited their compatibilities with SHJ solar cells [10, 14], obtaining efficiencies comparable to those acquired using ITO films. Furthermore, these films demonstrate excellent environmental stabilities including humidity resistance—an essential requirement for long-term durability in PV modules. Moreover, low-resistivity (ρ) SnO_2 films have been prepared at low temperatures via RPD using Sn source materials, and their effectiveness in SHJ solar cells have been independently demonstrated [12].

Despite its advantages, the widespread adoption of RPD in PV manufacturing remains limited mainly because the number of equipment manufacturers offering RPD systems is relatively small. In contrast, magnetron sputtering (MS) is extensively implemented in industry, supported by a broad range of equipment suppliers and well-established infrastructure for large-area processing. However, SnO_2 films deposited by MS at low temperatures ($\leq 200°C$) typically suffer from higher resistivities (~$3–5 \times 10^{-3}\,\Omega$cm) and undesired crystallization, which degrade their performances as TCOs [18–29]. Bombardment by high-energy negative O ions (O^-), which originate from the negatively biased target and are accelerated toward the growing film surface at energies of several hundred eV, is considered to be a major issue in MS [29]. These ions introduce O interstitials (O_i), which substantially deteriorate the electrical properties of SnO_2 as compared with the cases of ZnO or In_2O_3, owing to the inherently higher O ion flux during SnO_2 sputtering [30, 31].

To address these limitations, this study explores the development of highly conductive MS-deposited SnO_2 (MS-SnO_2) thin films via radio-frequency (RF) MS. Compared with conventional direct current sputtering, RF sputtering generally operates at lower target voltages, potentially suppressing the generation of high-energy O ions and reducing defect formation. Additionally, in some configurations, a lateral substrate offset from the erosion area is incorporated to spatially mitigate direct ion bombardment during growth. This study focuses on the correlation between amorphization and electrical performances of MS-SnO_2 films, which is critical for optimizing low-temperature processability. We systematically investigate the effects of substrate temperature, RF power, and water-related ambient conditions on film structure and transport properties. In particular, the influence of H arising from the residual water vapor in deposition chambers on carrier concentration (N) and mobility (μ) is analyzed. These investigations, along with comparative analysis with RPD-derived SnO_2 films, provide key insights into the conditions needed to enhance N and μ in MS-SnO_2 films. Accordingly, our optimized MS-SnO_2 films achieve resistivities as low as $1.2 \times 10^{-3}\,\Omega$cm, approaching the best values obtained by RPD ($9.3 \times 10^{-4}\,\Omega$cm), and exhibit the lowest ρ reported to date for SnO_2 films prepared by MS at the low process temperatures. SHJ solar cells employing MS-SnO_2 on both front and rear sides demonstrate a conversion efficiency of 21.1%, only slightly below those of RPD-derived SnO_2-based cells (22.5%) and ITO-based cells (22.1%). These results highlight the potential of MS-SnO_2 as a high-performance In-free TCO achievable by traditional sputtering, offering a practical pathway for the integration of MS-SnO_2 into scalable PV technologies.

2 | Methods

2.1 | Preparation of SnO_2 TCO Thin Films

SnO_2 thin films were deposited using two different RF-MS systems, referred to as MS-A and MS-B, to examine the influences of deposition conditions on the structural and electrical properties of TCO films. All films were deposited without substrate heating (substrate temperature $< 50°C$) and had thicknesses of approximately 100 nm for characterization. Post-deposition annealing was conducted under a N_2 atmosphere at 200°C for 30 min and at 250°C for 30 min for MS-B and MS-A films, respectively.

The first system, namely, MS-A, was a four-source RF sputtering system (Cannon Anelva, EB1100) equipped with a low-vacuum load-lock chamber evacuated using a rotary pump and main deposition chamber evacuated by a combination of turbo molecular and rotary pumps, realizing a base pressure below 3×10^{-5} Pa. To minimize the influence of residual water vapor and ensure reproducible deposition conditions, the substrate tray was transferred to the deposition chamber and held until the pressure decreased below 1×10^{-4} Pa, followed by a one-minute pre-sputtering step. During deposition, the total working pressure was maintained at 0.3 Pa, and O_2 flow ratio was defined as $r(O_2) = O_2$ flow rate/ (O_2 flow rate + Ar flow rate), which was varied to control oxygen stoichiometry. To avoid direct exposure to high-energy particle flux near the erosion area, all MS-A films were deposited on the substrates laterally offset by 110 mm and vertically positioned 150 mm above the target while continuously rotating to ensure uniform thickness. This substrate configuration was based on our previous

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study, which revealed that high-energy bombardment directly above the erosion area promoted crystallization even for films deposited without substrate heating [32]. Therefore, substrate placement outside this region was a critical design consideration for retaining the amorphous nature of the films. Deposition rates were approximately 6.0, 8.5, and 11 nm min^{-1} at RF powers of 100, 150, and 200 W, respectively. The second system, that is, MS-B, was a high-vacuum RF-MS system (Eiko Engineering) equipped with a turbo molecular pump and rotary pump for both load-lock and deposition chambers. A three-inch SnO_2 ceramic target was tilted at 60° relative to the substrate plane, with a target-to-substrate distance of approximately 160 mm [33]. The substrate holder was continuously rotated during deposition to guarantee homogeneous thickness. Films were deposited without intentional substrate heating. Base pressure of the deposition chamber was below 3×10^{-6} Pa, and total working pressure during deposition was maintained at 0.5 Pa. Effect of water vapor on the TCO properties was systematically analyzed by varying the partial pressure of H_2O (P_{H2O}) from 1×10^{-5} to 1×10^{-3} Pa using a variable leak valve [2]. Under these conditions, the deposition rate was approximately 2.7 nm min^{-1} at a RF power of 100 W.

For comparison, reference a-SnO_2 films were also prepared by RPD (Sumitomo Heavy Industry) [17, 34], following the procedure described in our previous study [16]. Films were deposited without intentional substrate heating (substrate temperature <60°C) under a total pressure of approximately 0.4 Pa using SnO_2 ceramic tablets as the source material. Base pressure prior to deposition was approximately 1×10^{-4} Pa, and deposition rate was approximately 130 nm min^{-1}. Post-deposition annealing was conducted under a N_2 atmosphere at 200°C for 30 min.

2.2 | Fabrication of SHJ Solar Cells With a-SnO_2 TCOs

SHJ solar cells were fabricated using *n*-type monocrystalline Si wafers (float-zone, ~2 Ωcm, <100> orientation, and 280 μm thickness) with randomly textured surfaces formed by KOH-based wet chemical etching. After RCA-based cleaning, intrinsic (*i*-type) and doped nc-Si:H or hydrogenated nanocrystalline silicon oxide (nc-SiO_x:H) layers were deposited on both wafer surfaces using plasma-enhanced chemical vapor deposition. In a typical rear-junction configuration, *n*-type nc-SiO_x:H was deposited on the front side and *p*-type nc-Si:H was deposited on the rear side. After the formation of the passivation and carrier-selective layers, a-SnO_2 films with thicknesses of ~75 nm were deposited as TCO layers by MS-A or RPD under the conditions explained in Section 2.1. ITO films were often used in our previous studies on SHJ cells [10, 14] and therefore employed as a benchmark TCO in this study. The films were fabricated using an RF-MS system (Ulvac, SBR-2306) with a commercial 4-inch ITO target (In_2O_3 + 10 wt.% SnO_2) at a pressure of 0.13 Pa using Ar as the process gas, with an $r(O_2)$ of 0.74%, and an RF power of 100 W. All TCO depositions were performed without intentional substrate heating. Post-deposition annealing was conducted at 160°C under a N_2 atmosphere for 120 min to recover potential degradation in carrier lifetime due to plasma damage during TCO deposition. For reference, the electrical properties of the ITO benchmark films after annealing at 160°C were evaluated. The ITO film exhibited a resistivity of 4.8×10^{-4} Ωcm, a carrier concentration of 2.4×10^{20} cm^{-3}, and an electron mobility of 54 $cm^2 V^{-1} s^{-1}$. Although higher annealing temperatures (e.g., 200°C–250°C) have been used in experiments for RPD- and MS-SnO_2 films, we have confirmed that annealing at 160°C is sufficient to induce structural relaxation in a-SnO_2 films and notably improves the electrical properties of these films. Finally, Ag electrodes were deposited on the TCO layers by MS via a shadow mask. All devices exhibited a monofacial architecture with a full-area Ag back electrode. The cell active area was 1 cm^2. In the front grid structure, the line widths and pitches of Ag fingers were 100 μm and 2.1 mm, respectively.

2.3 | Characterizations of TCOs and Solar Cells

Thicknesses of the TCO films were characterized by spectroscopic ellipsometry (J.A. Woollam, M-2000). Film structure was analyzed using out-of-plane grazing incidence X-ray diffraction (XRD) (Rigaku, SmartLab) to assess the presence of amorphous or nanocrystalline phases, and substrate surface was irradiated at an incident angle (ω) of 0.5° while scanning along only 2θ axis. Transmission electron microscopy (TEM) was also employed to evaluate grain structures and phase distributions for some samples. Optical properties of films on glass were measured using a spectrophotometer (Hitachi, U4000). Moreover, ρ, N, and μ were characterized via Hall effect measurements using the van der Pauw configuration (Toyo, ResiTest 8300). Work functions of TCO films were measured using a Kelvin prove equipment in vacuum (KP Technology, UHVKP020). H concentrations in the SnO_2 thin films were examined by secondary ion mass spectrometry (SIMS), which was performed at foundation for promotion of material science and technology of Japan. Reference sample consisting of a SnO_2 thin film implanted with H ions was used for calibration.

Current density–voltage (J–V) characteristics of the SHJ solar cell were evaluated using a class AAA solar simulator (Wacom, WXS-50S-L2) under standard test conditions (100 mW cm^{-2} and 25°C), and short-circuit current density (J_{SC}), open-circuit voltage (V_{OC}), fill factor (FF), and conversion efficiency (η) were evaluated. The cell area (1 cm^2) was designated by black shadow masks during the measurement. External quantum efficiency (EQE) spectra were measured by a Bunkou-Keiki, CEP-25BSX setup. Transfer length method (TLM) was applied to characterize specific contact resistivities of the electron contacts TCO/(*n*)nc-SiO_x:H/(*i*)a-Si:H and hole contacts TCO/(*p*)nc-Si:H/(*i*)a-Si:H used in the SHJ cells. TLM pattern simultaneously formed on the SHJ cell samples and was characterized. Note that textured *p*-type Si wafers (float zone, ~2 Ωcm, B-doped, <100> oriented) were utilized for the TLM characterization of the hole contacts.

3 | Results and Discussion

3.1 | Structure–Property Relationships in MS-SnO_2 Films

In this section, deposition parameters are discussed as means to control amorphization and the resulting Sn–O network structures in MS-SnO_2 films. The extracted structure–property

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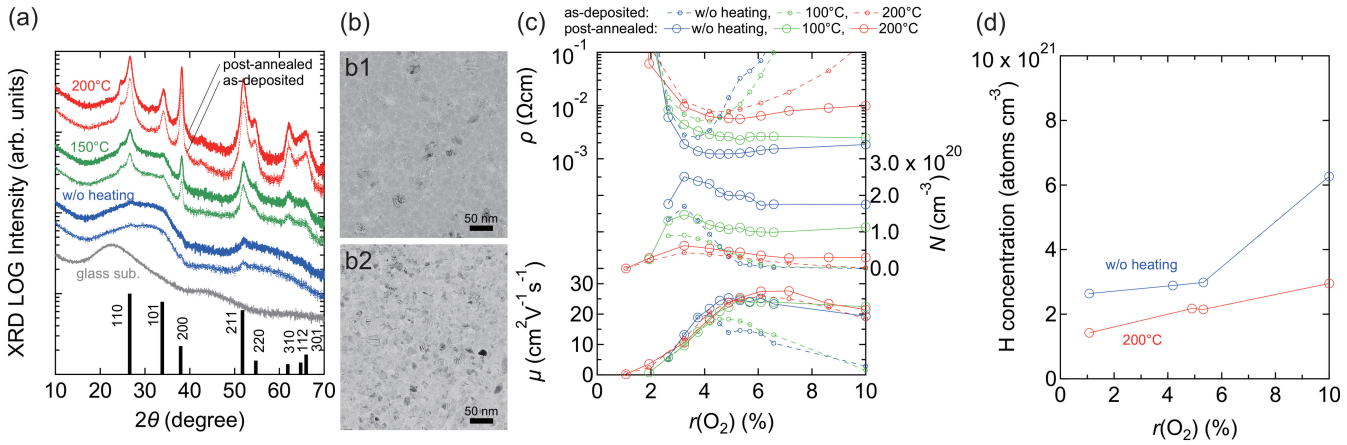


FIGURE 1 | (a) XRD patterns of the as-deposited and post-annealed SnO_2 films deposited at various substrate temperatures. Post-annealing was performed at 250°C. (b) Plane-view TEM images of post-annealed films deposited (b1) without substrate heating and (b2) at 200°C. (c) Dependence of resistivity (ρ), carrier concentration (N), and mobility (μ) on O_2 flow ratio ($r(O_2)$) for different deposition temperatures before and after post-annealing. (d) SIMS profiles of H concentration as a function of $r(O_2)$ for as-deposited films deposited without substrate heating and at 200°C.

relationships are expected to be relevant to low-temperature-processed SnO_2 thin films dominated by amorphous networks.

3.1.1 | Influence of Substrate Temperature

We investigated the effects of deposition temperature and $r(O_2)$ on the structural and electrical properties of the SnO_2 thin films deposited using MS-A. Films were deposited at three different substrate temperatures: room temperature (without heating), 100°C, and 200°C. For each temperature, $r(O_2)$ was varied. Figure 1a shows representative XRD patterns of as-deposited and post-annealed films, exhibiting high conductivity at each deposition temperature. For the films deposited without substrate heating, only weak diffraction peaks corresponding to rutile-type SnO_2 are observed, indicating that the films are predominantly amorphous. With an increase in the deposition temperature to 100°C and 200°C, both the intensities and sharpness of the peaks increase. At 200°C, the broad features characteristic of the amorphous phase disappears, suggesting complete crystallization of the film. Figure 1b depicts corresponding plane-view TEM images, supporting the XRD results. Film deposited without substrate heating exhibited an amorphous-dominant structure comprising nanocrystals embedded in an amorphous matrix. After post-annealing, partial crystallization occurs (b1); nevertheless, the amorphous phase remains dominant. In contrast, the film deposited at 200°C exhibits clearly defined crystalline grains (b2), verifying complete crystallization. In all cases, the grain size is relatively small, approximately 15–20 nm.

Figure 1c shows the variation in electrical properties as a function of $r(O_2)$. At $r(O_2) = 0$, the film deposited at room temperature shows a very high ρ (~1×10^5 Ωcm), and Hall effect measurements are not possible. However, with an increase in $r(O_2)$, both N and μ sharply increase, and the corresponding film exhibits a minimum ρ of approximately 2.5×10^{-3} Ωcm at $r(O_2) = \sim 4\%$ (N: 1.3×10^{20} cm^{-3} and μ: 19 cm^{2}V^{-1}s^{-1}). With further increases in $r(O_2)$, both N and μ gradually decrease and ρ increases again to above 10^{-1} Ωcm. Post-annealing at 250°C slightly affects the films deposited at low $r(O_2)$. On the other hand, for films deposited at $r(O_2) \gtrsim 4\%$, post-annealing significantly increases both N and μ, and a minimum ρ of approximately 1.2×10^{-3} Ωcm is achieved at $r(O_2) \approx 5\%$ (N: 2.1×10^{20} cm^{-3} and μ: 24 cm^{2}V^{-1}s^{-1}). Even at higher $r(O_2)$, low ρ is maintained due to the enhanced electrical properties. Similar trends in $r(O_2)$ dependence are observed for the films deposited at elevated temperatures (100°C and 200°C); however, the variations in N and μ with respect to $r(O_2)$ are more gradual in the as-deposited state, and the range of $r(O_2)$ resulting in minimum ρ is broad. Films deposited at high temperatures exhibit properties closer to those of the post-annealed films. Notably, minimum ρ values of the as-deposited films are approximately 2.5×10^{-3} Ωcm for room-temperature deposition, 5.1×10^{-3} Ωcm for 100°C deposition, and 7.8×10^{-3} Ωcm for 200°C deposition. After 250°C post-annealing, ρ decreases to 1.2×10^{-3}, 2.2×10^{-3}, and 5.6×10^{-3} Ωcm. These results reveal that lower N causes a degradation in electrical performance with an increase in the deposition temperature. Finally, Figure 1d depicts H concentrations in the films for the unheated and 200°C deposition conditions. For the unheated films, the H content is $3–6 \times 10^{21}$ cm^{-3} and increases with an increase in $r(O_2)$. In contrast, for the films deposited at 200°C, although H concentration also increases with an increase in $r(O_2)$, the absolute levels are lower ($1–3 \times 10^{21}$ cm^{-3}), indicating a slight reduction in H incorporation owing to elevated substrate temperatures.

The abovementioned results reveal that the electrical properties of the SnO_2 films are primarily governed by the degree of amorphization and the associated local Sn–O network structure, whereas H incorporation plays a secondary role. High conductivity was realized for amorphous-dominant films deposited at room temperature via appropriate O_2 flow control and post-annealing, demonstrating that high N and μ can be achieved even in the absence of full crystallinity. Moreover, the higher N observed in amorphous films compared with crystalline films deposited at elevated temperatures indicates that the amorphous structure itself favorably contributes to carrier generation. This contrast implies that the origin of carrier generation in amorphous and crystalline SnO_2 is fundamentally different. In crystalline films, carrier generation is conventionally attributed to point defects such as O vacancies, which may be less abundant

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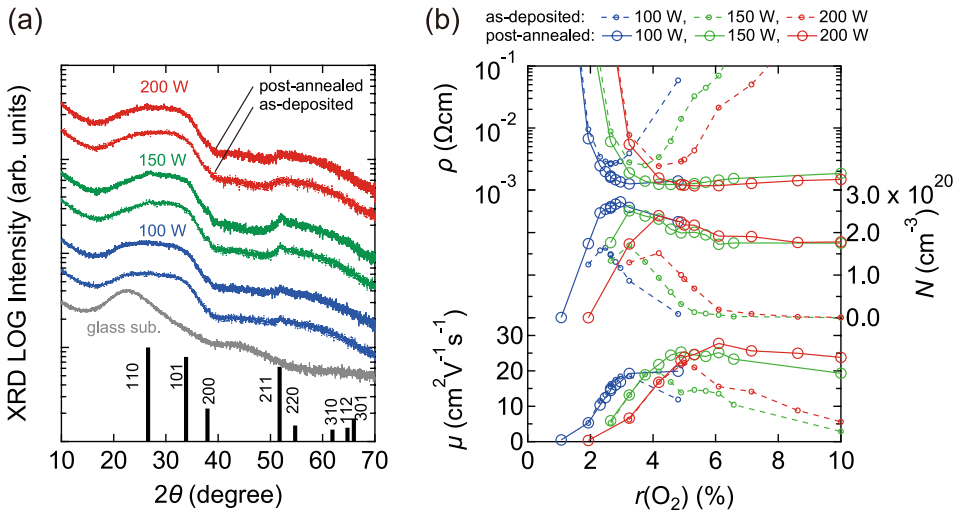


FIGURE 2 | (a) XRD patterns of as-deposited and post-annealed SnO_2 films deposited at different RF powers (100, 150, and 200 W) without substrate heating. (b) Dependences of resistivity (ρ), carrier concentration (N), and mobility (μ) on O_2 flow ratio ($r(O_2)$) for each RF power condition before and after post-annealing.

or less effective at deposition temperatures of ~200°C under moderate O_2 partial pressures. In contrast, amorphous SnO_2 possesses a disordered Sn–O network that enables a donor-active structural motifs inaccessible in the crystalline phase. These include not only under-coordinated Sn—O polyhedra but also peroxide-like O—O units that can emerge even under relatively O-rich conditions. Recent theoretical studies suggest that such peroxide-like configurations can induce donor-like electronic states characterized by spatially extended electron density near the conduction band edge, originating from network-driven electronic reconstruction rather than isolated point defects [35–38].

At extremely low $r(O_2)$, the formation of SnO-like local structures leads to low N and μ, which is unfavorable for TCO applications [16]. With an increase in $r(O_2)$, the local structure evolves toward SnO_2-like coordination, stabilization donor-active amorphous networks composed of under-coordinated Sn—O polyhedra and, potentially, peroxide-like O—O units. When oxygen incorporation exceeds this favorable regime, excessive distortion the amorphous Sn—O network introduces compensating configurations such as O interstitials or highly distorted polyhedra, which suppress donor activity and thus increase resistivity. Post-annealing can partially reverse these effects by promoting network relaxation and rearrangement, recovering high N, particularly in films grown at high $r(O_2)$. In addition to these intrinsic structural factors, H incorporation appears to play a non-negligible but secondary role in modulating the electrical properties. SIMS analysis revealed that a substantial amount of H ($>1\times10^{21}$ atoms cm^{-3}), primarily originating from residual water vapor, remained in the films, especially under low-temperature deposition conditions that favor amorphous structures. Rather than acting as a direct shallow donor, H is considered to indirectly influence carrier generation by relaxing the amorphous Sn—O network, passivating deep-level defects, and stabilizing donor-active configurations. This interpretation is consistent with recent theoretical insights into the functions of H introduction in amorphous oxide semiconductors [39–42]. However, although the H content monotonically increased with an increase in $r(O_2)$, the electrical conductivity demonstrated a maximum at moderate $r(O_2)$ and declined at higher $r(O_2)$ values. This behavior suggests that H incorporation alone does not guarantee high conductivity, reinforcing that the primary determinant is the local Sn—O network structure. At high $r(O_2)$, H is likely incorporated as hydroxyl (OH) species, which introduce deep-trap states and reduce both N and μ [39, 41, 42] (details are provided in Section 3.1.3). Overall, these results demonstrate that a moderately O_2-deficient growth environment is optimal for achieving high conductivity, as it stabilizes favorable donor-active amorphous Sn—O network structures while suppressing compensating defects and deep-level traps.

3.1.2 | Influence of RF Power

We investigated the variations in the structural and electrical properties of SnO_2 thin films with respect to RF power. Films were deposited without substrate heating, and RF powers of 100, 150, and 200 W were employed. For each RF power, $r(O_2)$ was systematically varied during deposition. The purposes of this analysis were to examine whether RF power affects the crystallinity of the films deposited without substrate heating and to verify the potential to obtain highly conductive amorphous films by optimizing RF power. Figure 2a shows representative XRD patterns of the films exhibiting high conductivity under each RF power condition. With variation in RF power, no significant changes are observed in the XRD patterns. Even at the lowest power of 100 W, diffraction peaks attributed to rutile-type SnO_2 are still slightly visible, indicating that completely amorphous films were not obtained. Furthermore, post-annealing did not cause any notable changes in the XRD patterns, indicating that RF power does not strongly influence crystallization under the present deposition conditions. Figure 2b depicts the electrical properties of the films obtained at different RF powers. With variation in $r(O_2)$, the O_2 ratio yielding the lowest ρ slightly shifts toward the higher O_2 side with an increase in the RF power. However, no considerable differences are observed in the overall ρ trends compared with those shown in Figure 1a. The minimum ρ are 2.4–2.6$\times10^{-3}$ Ωcm in the as-deposited state and 1.2×10^{-3} Ωcm after post-annealing, regardless

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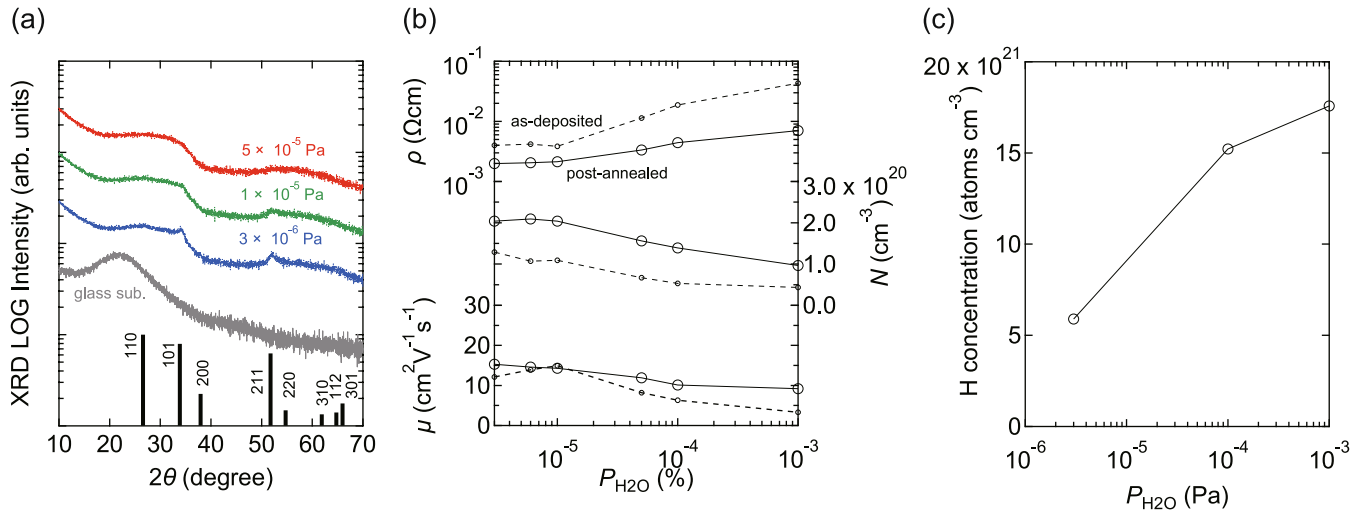


FIGURE 3 | (a) XRD patterns of the as-deposited SnO_2 films deposited under various water vapor partial pressures ($P_{H2O} = 3\times10^{-6}$–1×10^{-3} Pa) using MS-B without substrate heating. (b) Electrical properties (ρ, N, and μ) of the films with the lowest resistivities obtained under each water vapor condition. (c) H concentrations in the corresponding films as functions of water vapor partial pressure.

of RF power. In contrast, N and μ show systematic trends with respect to RF power: with a decrease in the RF power, N slightly increases, whereas μ decreases.

Although the XRD patterns indicate that completely amorphous structures were not achieved even at low RF power, increasing RF power also did not promote crystallization, and the amorphous phase remained dominant. This behavior can be attributed to the offset placement of the substrate from the erosion area of the target, which suppresses the influences of high-energy incident particles even at high RF power and helps retain an amorphous-like structure [32]. Moreover, even films classified as amorphous by XRD contain locally distributed nanocrystals (Figure 1b), suggesting that similar nanoscale structural features may be present in the films discussed in this section.

Next, the effects of RF power on N and μ are discussed. With an increase in the RF power, N decreased, whereas μ increased. This behavior is primarily ascribed to the suppression of scattering by charged donor-related electronic states: the reduction in the density of donor-active configurations responsible for free carriers probably reduces Coulombic scattering centers and thus enhances μ. The decrease in N at higher powers may be due to suppressed formation of donor-like structural motifs such as under-coordinated Sn–O polyhedra. Additionally, differences in H incorporation could indirectly contribute to the decrease in N. Specifically, higher RF power results in a higher deposition rate, which may reduce the relative introduction of H from residual water vapor in the chamber. As discussed in Section 3.1.1, H is unlikely to function as a direct shallow donor in these a-SnO_2 films; instead, its role is considered to be indirect. H may promote carrier generation by stabilizing donor-active structural configurations or by relaxing local structural distortions in the amorphous network. Therefore, reduced H incorporation at higher RF power likely limits these beneficial structural effects, leading to lower N. Importantly, despite the reduction in N, the observed high μ indicates that no significant carrier-trapping defects are produced. This supports the conclusion that the fundamental nature of the charge-transport mechanism remains similar across different RF power conditions, and that the observed trends mainly arise from variations in the density of donor-related electronic states and scattering effects rather than from degradation of film quality.

3.1.3 | Influence of Water Vapor Introduction

We investigated the impacts of intentional water vapor introduction during film deposition on the structural and electrical properties of SnO_2 films. This analysis was conducted to explore a method for promoting film amorphization and gain deeper insights into the effects of residual water vapor pressure in the deposition chamber discussed in the previous sections. For film deposition, MS-B was used instead of MS-A. RF power was fixed at 100 W, no substrate heating was applied, and $r(O_2)$ was varied over a wide range. P_{H2O} was gradually increased from 3×10^{-6} Pa (i.e., without intentional water vapor introduction) to 1×10^{-3} Pa. Figure 3a shows representative XRD patterns of the as-deposited films exhibiting high conductivities under various P_{H2O} and $r(O_2)$ values. For the film without intentional water vapor introduction, relatively strong 101 and 211 diffraction peaks are observed as compared with those depicted in Figures 1a and 2a, indicating a predominantly amorphous mixed-phase structure containing some crystalline components. In contrast, an increase in P_{H2O} gradually suppressed the intensities of the diffraction peaks. At $P_{H2O} \geq 5\times10^{-5}$ Pa, the diffraction peaks completely disappear, demonstrating the formation of a fully amorphous structure. Figure 3b shows the electrical properties of the films with the lowest ρ values under each water vapor condition. Up to a P_{H2O} of 1×10^{-5} Pa, no significant changes in electrical properties are noticed. However, at $P_{H2O} \geq 5\times10^{-5}$ Pa, both N and μ begin to gradually decrease, leading to a corresponding increase in ρ. The optimal $r(O_2)$ for achieving the lowest ρ was 0%–0.125%, which remained nearly unchanged regardless of P_{H2O}. Furthermore, under higher-$r(O_2)$ conditions, further decreases in N and μ are observed with an increase in P_{H2O}, which sharply increases ρ (details are provided in Figure S1). This trend is consistent with the results obtained using MS-A (Figures 1c,d). Figure 3c depicts the H concentrations in the SnO_2 films exhibiting the lowest ρ values under each water vapor condition. Clear

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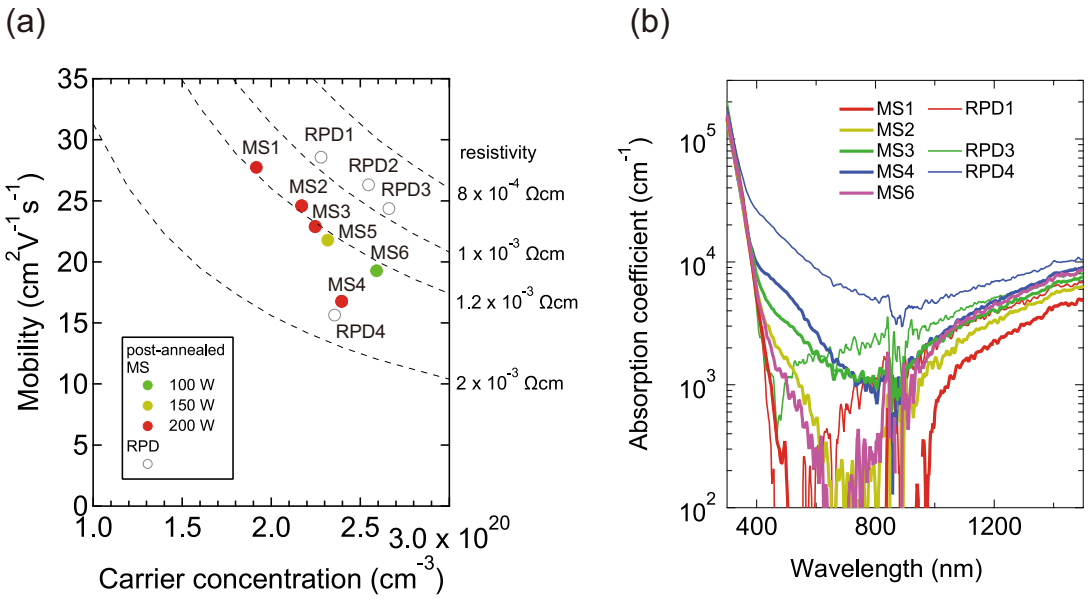


FIGURE 4 | (a) Hall mobilities versus carrier concentrations for the MS-SnO_2 films deposited at various O_2 flow ratios ($r(O_2)$) and RF powers (MS1–MS6) when compared with those for the RPD-SnO_2 films (RPD1–RPD3) acquired from reference [16]. (b) Absorption coefficient spectra of the same films. MS films exhibited high sub-bandgap absorptions, whereas RPD films exhibited minimal absorptions except under strongly reducing conditions (RPD4).

trends are observed: with an increase in P_{H2O}, the H concentration in the film also increases, suggesting effective incorporation of the H-containing species originating from water vapor during film growth. Interestingly, even under intentional water-free conditions, although MS-B demonstrates a better base pressure ($<3\times10^{-6}$ Pa), the H concentration is higher for MS-B than that observed for MS-A (Figure 1d).

These results clearly reveal that the introduction of water vapor promotes the amorphization of SnO_2 thin films and substantially influences both their structural and electrical properties. The disappearance of XRD peaks under P_{H2O} values of 5×10^{-5} Pa or higher implies that water vapor suppresses crystal nucleation on the growing film surface, thereby stabilizing the amorphous structure. Notably, the absence of significant change in N, μ, and ρ upon amorphization suggests that presence of nanocrystallites embedded in amorphous matrix does not largely affect charge transport. In contrast, when P_{H2O} exceeds 5×10^{-5} Pa, both N and μ decrease, leading to a gradual increase in ρ. This trend closely resembles the behavior observed under high $r(O_2)$ conditions (Section 3.1.1), supporting the conclusion that H concentrations above approximately 5×10^{21} cm^{-3} do not efficiently enhance carrier generation. Excessive H is likely incorporated as hydroxyl-related species, which introduce deep-trap states and thereby reduce both N and μ [39, 41, 42]. In addition, studies on In—O, Zn—Sn—O, and In—Ga—Zn—O systems have shown that high H concentrations can induce structural degradation, such as reduced film density and void formation, further enhancing carrier scattering and lowering μ [43–45]. Thus, both electronic effects (hydroxyl-related deep traps) and structural degradations contribute to the observed deterioration of electrical properties. Nevertheless, at H concentrations below approximately $3–5\times10^{21}$ cm^{-3}, H may still moderately contribute to carrier generation by passivating acceptor-like defects and stabilizing donor-active structural configurations in the amorphous Sn—O network. This indirect role of H is consistent with the discussion in Section 3.1.1, which indicates that H incorporation promotes high N specifically in amorphous films grown under moderately O_2 deficient conditions, where donor-like undercoordinated Sn—O polyhedra are more readily stabilized.

To further elucidate the role of H in amorphous SnO_2, it is essential to examine the relationship between H content and N under fully amorphous conditions with H concentrations below 1×10^{21} cm^{-3}. However, in the present MS process, reducing P_{H2O} or increasing the deposition temperature—both of which suppress H incorporation—also promote crystallization, making it difficult to decouple structural and electrical effects. Similar challenge is not unique to SnO_2, and similar difficulties have been encountered in In_2O_3 systems [43]. Compared with In_2O_3, SnO_2 exhibits greater chemical complexity, including multiple valence states (Sn^{4+}/Sn^{2+}) and variable O coordination environments, leading to more intricate interactions between H and the host network [38, 41]. In this context, the present identification of deposition conditions that stabilize the amorphous phase while maintaining relatively high conductivity represents a significant advancement, highlighting the importance of precise process control for tuning defect chemistry in amorphous TCOs.

3.2 | Application of MS-SnO_2 Films to SHJ Solar Cells

3.2.1 | Comparison of MS-SnO_2 Films With RPD-SnO_2 Films

Figure 4a shows the relationship between μ and N for the MS-SnO_2 films fabricated at 200 W using different $r(O_2)$ values (MS1–MS4) and MS-SnO_2 films deposited at 100, 150, and 200 W under respective optimized $r(O_2)$ conditions (MS6, MS5, and MS2). For comparison, data for the SnO_2 films prepared by

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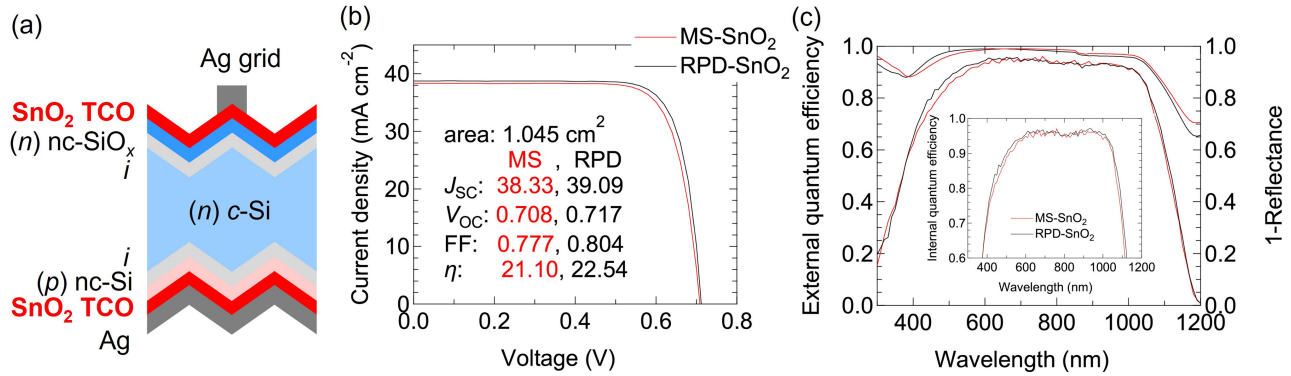


FIGURE 5 | (a) Schematic structure of the SHJ solar cells fabricated using MS-SnO_2 (MS2–MS4) or RPD-SnO_2 (RPD2) as front and rear electrodes. (b) Illuminated *J*–*V* characteristics (AM1.5G, 100 mW cm^{-2}) of cells with MS2 and RPD2 films. Inset: summary of photovoltaic parameters. (c) External quantum efficiency and reflectance spectra of the same cells. Inset: internal quantum efficiency spectra.

RPD (RPD1–RPD3) in a previous study [16] are also included. For both series, sample numbers increase with a decrease in the O_2 partial pressure during deposition, enabling systematic comparison of electrical properties as a function of $r(O_2)$. MS-SnO_2 films achieved minimum ρ values of approximately 1.2×10^{-3} Ωcm, which are considerably lower than the previously reported ρ range for these films (typically $3–5\times10^{-3}$ Ωcm) [18–29]. Although *N* values comparable with those of the RPD films were obtained by optimizing $r(O_2)$, the μ values of the MS films remained slightly lower. Consequently, the overall electrical conductivities of the MS films were slightly inferior to that of the most conductive RPD film ($\rho = 9.3\times10^{-4}$ Ωcm).

Figure 4b depicts the absorption coefficient spectra of the same films. For the MS-SnO_2 films deposited at 200 W (MS1–MS4), a substantial increase in sub-bandgap absorption around 500 nm is observed with a decrease in $r(O_2)$. This absorption is attributed to defect states associated with transitions involving valence band tail states, Sn^{2+} lone-pair electrons, O vacancies, or structural voids. The assignment of these defect states is based on our previous studies on amorphous SnO_x films, in which the O/Sn ratios, electronic states, and optical properties were systematically investigated using Rutherford backscattering spectrometry, hard X-ray photoelectron spectroscopy, and spectrophotometer [16]. This effect was particularly prominent in MS3 and MS4, which were deposited under low-$r(O_2)$ conditions and exhibited high *N* values. In contrast, RPD-SnO_2 films (RPD1 and RPD3) with similarly high *N* values exhibited minimal visible-range absorptions, and only RPD4, which was deposited at low $r(O_2)$, exhibited sub-bandgap absorption similar to those of MS films with even higher intensity.

These results suggest that sputtered films generally have a higher tendency to exhibit O-deficiency-related defects under low-$r(O_2)$ conditions. However, this trend is not absolute. For example, MS6, which was deposited at 100 W under even lower-$r(O_2)$ conditions than those for MS2–MS4, exhibited considerably lower sub-bandgap absorption. This initially counterintuitive result can be explained by considering the deposition dynamics: the low growth rate at 100 W allows more time for oxidation during film formation, which suppresses the generation of mid-gap defect states. Contrarily, although RPD is characterized by a significantly higher deposition rate than that of sputtering, corresponding films, except for RPD4, exhibit minimal sub-bandgap absorptions even at high *N* values. This can be ascribed to the highly reactive and oxidizing environment of RPD, which enables efficient Sn–O bond production and suppresses the incorporation of Sn^{2+} or O-deficiency-related defects despite the high deposition rate. These findings emphasize the critical influences of deposition dynamics, such as growth rate, ionization, precursor species, and their energies, on the resulting defect landscapes and optical qualities of SnO_2 films.

3.2.2 | SHJ Solar Cells With MS-SnO_2 and RPD-SnO_2 TCO Layers

We evaluated the impacts of MS-SnO_2 films (MS2–MS4) used as front and rear electrodes in SHJ solar cells on corresponding device performance. For comparison, we also fabricated solar cells using the RPD-SnO_2 film (RPD2) and our standard ITO film. Figure 5a shows the structures of the fabricated solar cells with the respective TCO films applied as front and rear electrodes, fabricated under identical design and process conditions. Figure 5b depicts the *J*–*V* characteristics under AM1.5G illumination for cells employing MS2, which exhibits both high conductivity and transparency among the MS-SnO_2 films, and RPD2 as the front and rear electrodes. Inset shows J_{SC}, V_{OC}, FF, and η for each cell. MS-SnO_2-based cell exhibits lower FF as compared with that of the RPD-SnO_2-based cell, indicating an impact of increased series resistance on FF. J_{SC} is also slightly lower as compared with that of the RPD-SnO_2 cell. To analyze the cause of J_{SC} reduction, the EQE spectra for cells using MS2 and RPD2 were acquired (Figure 5c). For the MS-SnO_2-based cell, a slight but noticeable decrease in spectral response is observed in the 400–600 and > 1000 nm wavelength ranges. The former is primarily attributed to reflection losses, owing to slight differences in TCO film thickness. To minimize the influence of optical interference, the inset of Figure 5c shows the internal quantum efficiency (IQE) spectra, derived from the EQE and reflectance spectra. Even in the IQE spectra, a modest reduction in spectral response is still evident for the MS-SnO_2-based cell. This is attributed to sub-bandgap absorption near the absorption edge, as well as free-carrier absorption in the MS film (Figure 4b). These optical losses contribute to the reduced quantum efficiency and J_{SC}. Overall, the use of MS-SnO_2 films results in a low conversion efficiency, primarily due to increased resistance and absorption-related losses.

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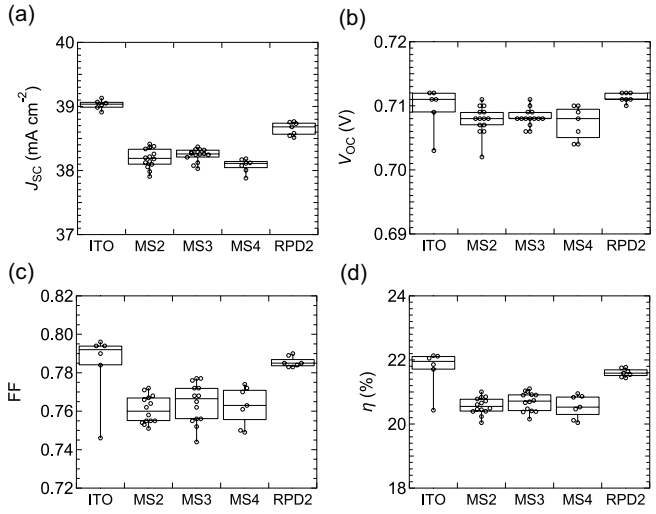


FIGURE 6 | (a) Short-circuit current density (J_{SC}), (b) open-circuit voltage (V_{OC}), (c) fill factor (FF), and (d) conversion efficiency (η) values of SHJ solar cells separately employing MS-SnO_2 (MS2–MS4), RPD-SnO_2 (RPD2), and ITO as front and rear electrodes.

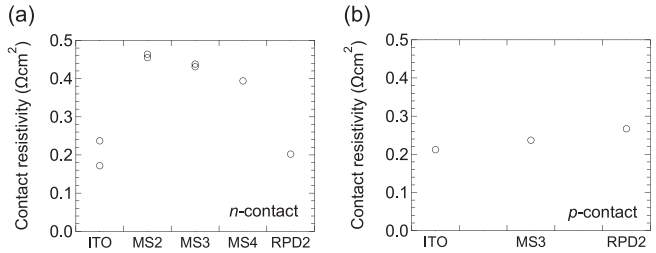


FIGURE 7 | Specific contact resistivities extracted using the transfer length method (TLM) for (a) electron-selective contacts formed between different TCO layers and nc-SiO_x:H (*n*)/*i* and (b) hole-selective contacts formed between different TCO layers and nc-Si:H (*p*)/*i*.

Thereafter, to verify whether these trends are consistent under several deposition conditions and perform a more comprehensive performance comparison, the performance metrics of SHJ solar cells separately using MS2, MS3, and MS4 (with different $r(O_2)$ values), RPD2, and ITO are determined and shown in Figure 6. Figure 6a–d depict the box plots of J_{SC}, V_{OC}, FF, and η for each cell. Additionally, Figure S2 shows relative performance values normalized to the ITO-based cell, facilitating direct comparison. In relative terms, the cells using RPD-SnO_2 and ITO demonstrate comparable performances, consistent with our previous findings [10]. On the other hand, MS-SnO_2-based cells exhibit ~2% lower J_{SC}, nearly the same V_{OC}, and 2%–3% lower FF values as compared with those of the RPD-SnO_2-based cells, leading to an overall efficiency reduction of up to ~5%. As previously discussed, the J_{SC} reduction is mainly ascribed to reflection and absorption losses in MS films.

For the FF reduction, the *J*–*V* curve in Figure 5b clearly indicates high series resistance. Possible factors contributing to this increased series resistance include (i) higher sheet resistance of the TCO film on textured substrates and (ii) high contact resistance with the carrier-selective layers. Although the sheet resistances of the TCO films on textured substrates are not directly measured in this study, they can be reasonably inferred based on film ρ and prior measurements reported in our previous study, where both ρ values and sheet resistances of the ITO and RPD-SnO_2 films on textured substrates are directly characterized [10]. Considering the ρ values obtained herein, although the estimated sheet resistances of the MS-SnO_2 films are higher than those of the ITO and RPD-SnO_2 films, they are not sufficiently large to account for the high series resistance observed in the *J*–*V* characteristics. This strongly suggests that contact resistance with the carrier-selective layers is the dominant factor contributing to this high series resistance. Therefore, we evaluated the contact resistances with both *n*-type nc-SiO_x:H and *p*-type nc-Si:H layers using TLM. Corresponding results are shown in Figure 7a,b. For the contact with the *n*-type layer, MS-SnO_2 exhibited significantly higher contact resistance as compared with those of other TCOs. In contrast, for the contact with the *p*-type layer, MS-SnO_2 showed slightly lower contact resistance than that of RPD-SnO_2.

High contact resistance at the MS-SnO_2/(*n*)nc-SiO_x:H interface emerges as a key factor limiting FFs and overall performance of the SHJ solar cells employing MS-SnO_2 as front TCO. This high contact resistance is partly attributed to hindered carrier transport caused by the formation of an interfacial energy barrier at the TCO/Si interface. A wide depletion region can predominantly form on the Si side, when the work function of TCO is substantially different from that of the Si-based layer, particularly when TCO possesses a higher work function and (*n*)nc-Si:H demonstrates relatively low *N*. This impedes current extraction and increases contact resistance. Our previous Kelvin probe measurements revealed that RPD-SnO_2 has a slightly higher work function than ITO by 32 meV [10], which induces expansion of the depletion layer and increases in contact resistance at the SnO_2/(*n*)Si interface. Similarly, the same measurements revealed that MS2, MS3, and MS4 have slightly higher work functions by 30, 37, and 34 meV, respectively, compared with RPD2. This is expected to further increase the interfacial energy barrier at the interface and likely contributes to the higher contact resistance observed in Figure 7a, although this factor alone may not fully account for the large difference compared with RPD-SnO_2. In contrast, the slightly lower contact resistance at the MS-SnO_2/(*p*)nc-Si:H interface suggests that the relatively high work function of MS-SnO_2 could be advantageous for hole transport in this configuration. In addition to the work function mismatch, sputtering-induced plasma damage to the thin carrier-selective Si layers is considered another important factor contributing to the high contact resistance at the MS-SnO_2/(*n*)nc-SiO_x:H interface. It has been widely reported that energetic ion bombardment during TCO sputtering can degrade the structural and electronic properties of nc-SiO_x:H and related contact layers, leading to increased contact resistance and reduced FF in SHJ solar cells. Recent studies have directly demonstrated that sputtered TCO deposition can deteriorate contact passivation and carrier extraction at the TCO/Si interface [46]. Given that MS inherently involves energetic plasma species, whereas RPD is a low-damage growth technique, plasma-induced degradation of the (*n*)nc-SiO_x:H layer may play a non-negligible role in the higher contact resistance observed for MS-SnO_2-based devices, in addition to the interfacial energy barrier effects. These considerations indicate that the high contact resistance originates from a combination of interfacial energy barrier effects and process-induced interfacial degradation.

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These findings underscore the importance of reducing contact resistance for improving both FF and overall device efficiency. Optimizing the carrier-selective layers themselves is a promising strategy in this regard. In our previous study, replacing a-Si:H with nc-SiO_x:H or nc-Si:H as the electron-selective layer considerably reduced contact resistance and improved FF due to enhanced doping efficiency and consequently improved current extraction [14]. Moreover, several reports have demonstrated that the combination of MS-SnO_2 and highly doped (*n*)nc-Si:H can achieve lower contact resistance and/or better FF [11, 13] than those observed in our current devices, indicating that optimizing *N* of the Si side remains a viable approach. On the TCO side, realizing low contact resistance requires a sufficiently high *N* for an appropriate work function alignment with the carrier-selective layers. Indeed, contact resistance decreases with an increase in *N* under low $r(O_2)$ (e.g., MS2–4), although the associated change in work function is relatively minor. In addition, there is a large difference in *n*-contact resistance between MS4 and RPD2, which show similar *N* values. This observation further supports the presence of interfacial degradation mechanisms discussed above. The slight lower V_{OC} (less than 1%) may be related to interfacial degradation induced during TCO deposition. Although low-power deposition (e.g., 100 W and MS6) may mitigate these drawbacks, yielding lower plasma damage and higher *N* with suppressed absorption, the deposition rate is impractically low due to RF sputtering with the offset-target configuration. In contrast, RPD represents a useful reference process, as it can achieve relatively high carrier concentration and transparency while mitigating plasma-induced damage (e.g., RPD2 and 3), together with a high deposition rate. Such characteristics contribute to reduced contact resistance without significantly compromising manufacturability. While advances in sputtering-based manufacturing approaches that enable low-damage deposition without sacrificing deposition rate may provide additional opportunities, the present results suggest that further progress in MS-SnO_2-based films should primarily rely on material-side innovations. Compositional engineering, such as forming ternary amorphous alloys [47], may offer an effective route to systematically tune the electronic properties of SnO_2. Unlike conventional aliovalent doping in crystalline films, alloying within an amorphous structure allows greater flexibility in controlling the work function and electron affinity while maintaining transparency and conductivity. Finally, it should be noted that several studies have recently demonstrated that high-efficiency, indium-saving SHJ solar cells with low contact resistance and high FF can be achieved when ITO or high-mobility In_2O_3:H-based layers are inserted between the silicon *n*-contact layer and the SnO_2-based TCO electrode by MS [11].

4 | Conclusions

This study demonstrates that magnetron-sputtered SnO_2 films can serve as promising In-free TCOs for SHJ solar cells when carefully optimized. By controlling substrate temperature, $r(O_2)$, and residual water vapor pressure, we achieved predominantly amorphous SnO_2 films with low ρ values (~1.2×10^{-3} Ωcm), attributed to the formation of shallow donor states in the disordered network and the reduced contribution of grain boundary scattering in amorphous-dominant films. H, introduced by the residual water vapor in the sputtering atmosphere, may play a secondary and indirect role by relaxing the amorphous Sn–O network or passivating deep-level defects, rather than acting as a direct donor. However, excessive H, particularly under high-water and high-O_2 partial pressures, degraded film quality owing to hydroxyl-related defects. Optimized MS-SnO_2 films afforded SHJ solar cells with a conversion efficiency of 21.1%, exhibiting good compatibility with standard SHJ device fabrication process, despite slightly lower performance compared with devices employing ITO or RPD-SnO_2 films. The performance limitation was mainly ascribed to high contact resistance at the MS-SnO_2/*n*-type nc-SiO_x:H interface. These findings indicate that high conductivity and compatibility with SHJ devices are achievable for a-SnO_2 films via MS. Future efforts should focus on interface engineering and work function control to reduce contact resistance and thereby close the efficiency gap with In_2O_3-based devices.

Author Contributions

Takashi Koida designed the experiments, performed the thin film fabrication and characterization (TCO), analyzed the data, and wrote the original manuscript. **Hitoshi Sai** led the solar cell fabrication and characterization and made contributions to the interpretation of the solar cell results and to the critical revision of the corresponding sections of the manuscript. **Takuya Matsui** supported the solar cell fabrication and characterization environment and made contributions to the interpretation of the solar cell results and to the critical revision of the corresponding sections of the manuscript.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Dependences of resistivity (ρ), carrier concentration (N), and mobility (μ) on O_2 flow ratio ($r(O_2)$) for the as-deposited and post-annealed films prepared under various water vapor partial pressures (Pa). **Figure S2:** Relative (a) short-circuit current density (J_{SC}), (b) open-circuit voltage (V_{OC}), (c) fill factor (FF), and (d) conversion efficiency (η) values of the SHJ solar cells separately employing MS-SnO_2 (MS2–MS4) and RPD-SnO_2 (RPD2) films versus those for the SHJ solar cells using ITO films as the front and rear electrodes.

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